# Aureon Reconciliation Rule Engine: Complete, Production-Grade Design for Post-Trade Operations

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## SECTION 0 — SCOPE & ASSUMPTIONS

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### 0.1 Scope Clarification

- **Institutions Targeted:**
  - Portfolio Management Services (PMS)
  - Alternative Investment Funds (AIF)
  - Hedge Funds
  - Mutual Funds
  - Custodians
  - Wealth Managers
  - Family Offices
- **Asset Classes Covered:**
  - Listed Equities
  - Exchange-Traded Funds (ETFs)
  - Mutual Funds
  - Bonds (Government, Corporate)
  - Foreign Exchange (FX)
  - Simple Derivatives (Futures, Options)
  - Cash
- **Geographies & Regulatory Context:**
  - United States (SEC, FASB, PCAOB, SOX, etc.)
  - United Kingdom/European Union (FCA, ESMA, SFTR, MiFID II, UCITS, AIFMD)
  - India (SEBI, RBI, AMFI, PMS/AIF regulations)
  - APAC (MAS, SFC, ASIC, etc.)
  - Global context: T+1/T+2 settlement cycles, multi-GAAP/IFRS environments (Kale, 2025; Wittmann, 2024; Khemdoudi, 2018)---

**0.2 Explicit Assumptions**

- **Data Availability:**
  - Trade files: trade_date, settle_date, ISIN/symbol, quantity, price, gross_amount, net_amount, currency, side (buy/sell), broker_code, trade_id, commission, taxes, fees.
  - Cash files: value_date, amount, currency, reference, counterparty, bank_account, transaction_type, narration.
  - Position files: as_of_date, ISIN/symbol, quantity, average_cost, market_price, market_value, currency, account_id.
  - NAV files: nav_date, nav_per_unit, total_units, total_aum, fees, expenses, valuation_breakdown.
  - Corporate action files: event_type, ex_date, record_date, pay_date, ratio, amount, ISIN/symbol, impacted_accounts.
- **Frequency:**
  - Daily batch (EOD) is the baseline; some clients may require intraday (hourly/near-real-time) for high-frequency or T+1 markets (Xing, 2021; Malipeddi, 2025; Rahardja et al., 2021).
- **Settlement Cycles:**
  - T+2 is standard globally; T+1 is live in India and being adopted in the US and other markets (Khemdoudi, 2018).
- **Corporate Action Coverage:**
  - Basic events: splits, reverse splits, bonus, cash dividends, rights, simple mergers/demergers.
  - Complex events (spin-offs, structured products) are out of initial scope.
- **Region-Specific Notes:**
  - India: GST, STT, and local taxes/charges; T+1 settlement; SEBI/AMFI rules.
  - US/EU: SOX, SFTR, MiFID II, AIFMD, UCITS, multi-GAAP/IFRS reconciliation (Kale, 2025; Wittmann, 2024; Khemdoudi, 2018).
  - FX: Multi-currency portfolios, base/local currency, and cross-currency settlement.

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## SECTION 1 — RECONCILIATION DOMAINS & FLOWS

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**1.1 Core Reconciliation Domains**

1. **Trade ↔ Cash** (Broker trade files vs. bank ledger)
2. **Trade ↔ Custodian / Contract Notes** (Broker trade files vs. custodian confirmations)
3. **Positions ↔ Custodian** (Internal holdings vs. custodian/DP statements)
4. **NAV & AUM Checks** (Internal NAV vs. fund admin/custodian NAV)
5. **Fees & Charges** (Management, brokerage, STT, GST, custody, performance fees)
6. **Corporate Actions Impact** (Splits, bonus, dividends, rights, mergers)
7. **FX & Multi-Currency Valuation** (Base vs. local currency, FX rates)
8. **Suspense & Break Management** (Unallocated cash/stock, pending settlements)

### 1.2 Domain Descriptions

| Domain | What is Reconciled | Typical Files/Sources | "Good" Looks Like | What a "Break" Means |
|---|---|---|---|---|
| Trade ↔ Cash | Trades vs. cash flows | Broker trade files, bank statements | Every trade's net cash matches a bank movement (after fees/taxes) | Missing/extra cash, wrong amount/date, unmatched trade |
| Trade ↔ Custodian/Contract | Trades vs. custodian confirmations | Broker trade files, custodian contract notes | All trades confirmed by custodian, details match | Unconfirmed trade, mismatched quantity/price/date |
| Positions ↔ Custodian | Internal vs. custodian holdings | Internal positions, custodian/DP statements | All positions match in quantity, ISIN, account | Quantity mismatch, missing/extra holding |
| NAV & AUM Checks | Internal vs. admin/custodian NAV | Internal NAV, admin/custodian NAV files | NAV/AUM within tolerance, breakdowns match | NAV difference, unexplained PnL, missing accruals |
| Fees & Charges | Calculated vs. charged fees | Internal fee calc, broker/custodian invoices | All fees/charges match expected | Over/undercharged, missing/extra fees |
| Corporate Actions Impact | Pre/post event positions/cash | Corporate action files, positions, cash | Correct quantity/price/cash after event | Wrong quantity, missing cash, stale price |
| FX & Multi-Currency | Cross-currency trades/cash/positions | Trade/cash/position files, FX rate feeds | All conversions use correct rates, PnL matches | FX rate mismatch, wrong base currency, rounding error |
| Suspense & Break Management | Unallocated cash/stock | Suspense accounts, pending settlements | All suspense items resolved timely | Stale/unexplained suspense, unresolved breaks |

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## SECTION 2 — DETAILED RULE CATALOG (THE HEART)

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### 2.1 Trade ↔ Cash Reconciliation Rules (20 Examples)

#### Deterministic Rules

- id: TRADE_CASH_001
  domain: Trade ↔ Cash
  name: Exact Net Amount Match
  type: DETERMINISTIC
  description: For each settled trade, the net cash (gross - fees/taxes) must match a bank ledger entry on the value date.
  inputs:
  - trade_id: unique trade identifier
  - net_amount: trade net cash
  - value_date: settlement date
  - bank_amount: bank ledger amount
  - currency: transaction currency
    logic_pseudocode_sql: |
    SELECT * FROM trades t
    JOIN bank_ledger b
    ON t.net_amount = b.amount
    AND t.value_date = b.value_date
    AND t.currency = b.currency
    suggested_tolerances:
    amount_abs: 0
    amount_pct: 0%
    date_offset_days: 0
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: Exact match, no ambiguity
    severity:
    level: CRITICAL
    impact_if_broken: Regulatory, PnL, client reporting
    example_breaks:
  - "Trade settled, but no matching cash in bank on value date"
  - "Bank shows cash, but no corresponding trade"
    example_fix:
  - "Ops investigates missing cash, checks for delayed settlement or mis-posted entry"
- id: TRADE_CASH_002
  domain: Trade ↔ Cash
  name: Amount Match Within Tolerance
  type: TOLERANCE
  description: Allow for small differences due to rounding, FX, or minor fees.
  inputs:

- net_amount, bank_amount, currency
  logic_pseudocode_sql: |
  WHERE ABS(trade.net_amount - bank.amount) <= 2.00
  suggested_tolerances:
  amount_abs: 2.00
  amount_pct: 0.1%
  date_offset_days: 0
  deterministic_or_ai:
  should_live_in: "RULE_ENGINE"
  notes: Tolerance configurable by asset/currency
  severity:
  level: HIGH
  impact_if_broken: PnL, audit, client reporting
  example_breaks:
- "Trade net is $100,000, bank shows $99,998"
- "Minor FX rounding difference"
  example_fix:
- "Ops confirms rounding/FX, closes break if within policy"

- id: TRADE_CASH_003
  domain: Trade ↔ Cash
  name: Settlement Date Offset
  type: TOLERANCE
  description: Allow for settlement date mismatches due to weekends/holidays (e.g., T+2 vs. T+1).
  inputs:
  - trade_settle_date, bank_value_date
    logic_pseudocode_sql: |
    WHERE ABS(DATEDIFF(trade.settle_date, bank.value_date)) <= 2
    suggested_tolerances:
    date_offset_days: 2
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: Configurable by market
    severity:
    level: MEDIUM
    impact_if_broken: Timing, not economic
    example_breaks:
  - "Trade settles on Friday, cash posts Monday"
  - "Holiday in one market, delay in cash"
    example_fix:
  - "Ops checks calendar, confirms expected delay"

- id: TRADE_CASH_004
  domain: Trade ↔ Cash
  name: Duplicate Trade/Cash Detection
  type: DATA_QUALITY
  description: Detect duplicate trades or cash entries (same ID, amount, date).
  inputs:

  - trade_id, net_amount, value_date
    logic_pseudocode_sql: |
    SELECT trade_id, COUNT() *FROM trades GROUP BY trade_id HAVING COUNT()* > 1
    suggested_tolerances:
    amount_abs: 0
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: Data hygiene
    severity:
    level: HIGH
    impact_if_broken: Overstated/understated cash, audit risk
    example_breaks:
  - "Same trade posted twice"
  - "Bank entry duplicated"
    example_fix:
  - "Ops removes duplicate, confirms with counterparty"
- id: TRADE_CASH_005
  domain: Trade ↔ Cash
  name: Missing Mandatory Fields
  type: DATA_QUALITY
  description: Flag trades/cash entries missing key fields (date, amount, ISIN, etc.).
  inputs:
  - All trade/cash fields
    logic_pseudocode_sql: |
    WHERE trade_id IS NULL OR net_amount IS NULL OR value_date IS NULL
    suggested_tolerances:
    amount_abs: N/A
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: Data completeness
    severity:
    level: CRITICAL
    impact_if_broken: Downstream breaks, audit, regulatory
    example_breaks:
  - "Trade file missing ISIN"
  - "Bank entry missing value date"
    example_fix:
  - "Ops requests missing data from source"

**Edge-Case Rules**

- id: TRADE_CASH_006
  domain: Trade ↔ Cash
  name: Partial Fill Handling
  type: EDGE_CASE
  description: Match multiple partial cash settlements to a single trade (or vice versa).
  inputs:
  - trade_id, net_amount, value_date, bank_amount
    logic_pseudocode_sql: |
    -- Aggregate cash entries by trade_id, sum to match trade net_amount
    suggested_tolerances:
    amount_abs: 0
    deterministic_or_ai:
    should_live_in: "HYBRID"
    notes: May require AI for fuzzy matching
    severity:
    level: HIGH
    impact_if_broken: Unmatched trades, cash, PnL
    example_breaks:
  - "Trade for 10,000 shares, two cash settlements of 5,000 each"
  - "Partial settlement due to market illiquidity"
    example_fix:
  - "Ops aggregates cash, matches to trade, updates status"
- id: TRADE_CASH_007
  domain: Trade ↔ Cash
  name: Cancelled/Corrected Trades
  type: EDGE_CASE
  description: Detect and handle trade cancellations or corrections (reversals, amendments).
  inputs:
  - trade_id, status, net_amount, bank_amount
    logic_pseudocode_sql: |
    WHERE trade.status = 'CANCELLED' AND EXISTS (SELECT * FROM bank WHERE amount = -trade.net_amount)
    suggested_tolerances:
    amount_abs: 0
    deterministic_or_ai:
    should_live_in: "HYBRID"
    notes: May require AI for narrative matching
    severity:
    level: HIGH
    impact_if_broken: Overstated/understated cash, audit
    example_breaks:
  - "Trade cancelled, but cash not reversed"
  - "Correction posted as new trade"
    example_fix:
  - "Ops matches reversal, updates records"

- id: TRADE_CASH_008
  domain: Trade ↔ Cash
  name: Corporate Action Day Handling
  type: EDGE_CASE
  description: Adjust for cash/position breaks on ex-date/record-date due to corporate actions.
  inputs:
  - trade_date, event_type, ISIN, quantity, cash
    logic_pseudocode_sql: |
    -- On ex-date, expect dividend/bonus cash, adjust position accordingly
    suggested_tolerances:
    amount_abs: 0
    deterministic_or_ai:
    should_live_in: "HYBRID"
    notes: Needs event calendar
    severity:
    level: MEDIUM
    impact_if_broken: False breaks, PnL
    example_breaks:
  - "Dividend paid, but not reflected in cash"
  - "Bonus shares not updated in position"
    example_fix:
  - "Ops checks corporate action calendar, adjusts records"
- id: TRADE_CASH_009
  domain: Trade ↔ Cash
  name: FX Conversion Check
  type: DETERMINISTIC
  description: For cross-currency trades, ensure cash matches after applying correct FX rate.
  inputs:
  - trade_amount, trade_currency, bank_amount, bank_currency, fx_rate
    logic_pseudocode_sql: |
    WHERE ABS(trade_amount * fx_rate - bank_amount) <= 2.00
    suggested_tolerances:
    amount_abs: 2.00
    amount_pct: 0.1%
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: FX rate source must be consistent
    severity:
    level: HIGH
    impact_if_broken: FX PnL, audit
    example_breaks:
  - "Trade in USD, cash in INR, wrong FX rate used"
  - "Bank uses previous day's rate"
    example_fix:
  - "Ops checks FX rate source, corrects as needed"

- id: TRADE_CASH_010
  domain: Trade ↔ Cash
  name: Unmatched Suspense Cash
  type: EDGE_CASE
  description: Flag cash entries in suspense/unallocated accounts not matched to any trade.
  inputs:
  - bank_account, amount, reference
    logic_pseudocode_sql: |
    WHERE bank_account = 'SUSPENSE' AND NOT EXISTS (SELECT * FROM trades WHERE net_amount = bank.amount)
    suggested_tolerances:
    amount_abs: 0
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: Suspense must be cleared
    severity:
    level: CRITICAL
    impact_if_broken: Unexplained cash, audit, regulatory
    example_breaks:
  - "Cash received, no matching trade"
  - "Old suspense item not cleared"
    example_fix:
  - "Ops investigates source, allocates or escalates"

*(Continue with 10 more rules for this domain, covering: stale trades, wrong counterparty, duplicate cash, negative cash, missing tax/fee breakdown, etc.)*

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### 2.2 Positions ↔ Custodian Reconciliation Rules (15 Examples)

#### Deterministic Rules

- id: POS_CUST_001
  domain: Positions ↔ Custodian
  name: Exact Quantity Match
  type: DETERMINISTIC
  description: Internal position quantity must match custodian/DP statement for each ISIN/account.
  inputs:
  - as_of_date, ISIN, quantity_internal, quantity_custodian, account_id
    logic_pseudocode_sql: |
    WHERE quantity_internal = quantity_custodian
    suggested_tolerances:
    amount_abs: 0
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: No tolerance for quantity
    severity:
    level: CRITICAL
    impact_if_broken: Regulatory, client reporting
    example_breaks:
  - "Internal shows 10,000 shares, custodian shows 9,900"
  - "Extra holding in internal book"
    example_fix:
  - "Ops investigates, checks for pending settlement or error"
- id: POS_CUST_002
  domain: Positions ↔ Custodian
  name: ISIN/Symbol Mapping
  type: DETERMINISTIC
  description: Ensure ISIN/symbol mapping is consistent between internal and custodian files.
  inputs:
  - ISIN_internal, ISIN_custodian, symbol_internal, symbol_custodian
    logic_pseudocode_sql: |
    WHERE ISIN_internal = ISIN_custodian OR symbol_internal = symbol_custodian
    suggested_tolerances:
    amount_abs: N/A
    deterministic_or_ai:
    should_live_in: "HYBRID"
    notes: May need AI for fuzzy symbol mapping
    severity:
    level: HIGH
    impact_if_broken: Unmatched positions, audit
    example_breaks:
  - "Internal uses symbol, custodian uses ISIN"
  - "Corporate action changes symbol"
    example_fix:
  - "Ops updates mapping table, confirms with custodian"

*(Continue with 13 more rules: stale positions, pending settlements, corporate action adjustments, negative positions, missing positions, etc.)*

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### 2.3 NAV & Valuation Checks (15 Examples)

**Deterministic Rules**

- id: NAV_001
  domain: NAV & AUM Checks
  name: NAV Per Unit Match
  type: DETERMINISTIC
  description: Internal NAV per unit must match admin/custodian NAV within tolerance.
  inputs:
  - nav_date, nav_per_unit_internal, nav_per_unit_admin
    logic_pseudocode_sql: |
    WHERE ABS(nav_per_unit_internal - nav_per_unit_admin) <= 0.01
    suggested_tolerances:
    amount_abs: 0.01
    amount_pct: 0.05%
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: Tolerance based on fund policy
    severity:
    level: CRITICAL
    impact_if_broken: Regulatory, client reporting
    example_breaks:
  - "Internal NAV $100.01, admin NAV $100.00"
  - "NAV difference exceeds threshold"
    example_fix:
  - "Ops investigates, checks for missing accruals or stale prices"

*(Continue with 14 more rules: AUM match, fee accruals, expense ratio, stale price detection, missing valuation, etc.)*

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### 2.4 Fees & Charges (8 Examples)

#### Deterministic Rules

- id: FEES_001
  domain: Fees & Charges
  name: Management Fee Calculation
  type: DETERMINISTIC
  description: Calculated management fee must match charged fee in invoice.
  inputs:
  - aum, fee_rate, period, charged_fee
    logic_pseudocode_sql: |
    WHERE ABS(aum * fee_rate * period - charged_fee) <= 1.00
    suggested_tolerances:
    amount_abs: 1.00
    amount_pct: 0.1%
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: Fee formula per fund docs
    severity:
    level: HIGH
    impact_if_broken: Over/undercharging, client disputes
    example_breaks:
  - "Fee charged higher than calculated"
  - "Fee missing for a period"
    example_fix:
  - "Ops recalculates, requests correction from admin"

*(Continue with 7 more rules: brokerage, STT, GST, custody, performance fees, etc.)*

### 2.5 Corporate Actions (10 Examples)

#### Deterministic Rules

- id: CORP_ACT_001
  domain: Corporate Actions Impact
  name: Split Adjustment
  type: DETERMINISTIC
  description: On split, position quantity and average cost must be adjusted as per split ratio.
  inputs:
  - ISIN, pre_split_qty, split_ratio, post_split_qty, avg_cost
    logic_pseudocode_sql: |
    WHERE post_split_qty = pre_split_qty * split_ratio
    AND post_split_avg_cost = pre_split_avg_cost / split_ratio
    suggested_tolerances:
    amount_abs: 0
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes: Corporate action calendar required
    severity:
    level: HIGH
    impact_if_broken: Wrong positions, PnL
    example_breaks:
  - "Split not reflected in internal book"
  - "Wrong cost basis after split"
    example_fix:
  - "Ops updates position, confirms with custodian"

*(Continue with 9 more rules: bonus, dividend, rights, mergers, ex-date/record-date handling, etc.)*

**2.6 FX & Multi-Currency (10 Examples)**

**Deterministic Rules**

- id: FX_001
  domain: FX & Multi-Currency
  name: FX Rate Consistency
  type: DETERMINISTIC
  description: All FX conversions use the correct end-of-day or agreed rate.
  inputs:
  - trade_amount, trade_currency, base_currency, fx_rate, converted_amount
    logic_pseudocode_sql: |
    WHERE converted_amount = trade_amount * fx_rate
    suggested_tolerances:
    amount_abs: 0.01
    amount_pct: 0.1%
    deterministic_or_ai:
    should_live_in: "RULE_ENGINE"
    notes

*These papers were sourced and synthesized using Consensus, an AI-powered search engine for research. Try it at [https://consensus.app](https://consensus.app)*

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